

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: WXIMPACTBENCH | Context: Gran Chaco Tri-Border Emergency Response Zone

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's entire input corpus is English-language formal newspaper prose from three Canadian newspapers (La Presse, La Patrie, Montreal Gazette) accessed via McGill University Library and the Bibliothèque nationale du Québec [Q97]. Dataset analysis confirmed every one of 31 sampled examples is English Canadian newspaper content; no Spanish, no social media, no indigenous-language toponyms, no Southern Cone references. The deployment requires Rio Platense Spanish social media and local news with Guaraní/Quechua/Wichí lexical borrowing [WEB-7, WEB-9, WEB-10]. The benchmark explicitly acknowledges potential biases in 'underrepresented historical events and linguistic variations' [Q90] but does not extend to Spanish-language or Southern Cone content. The regional context confirms no Spanish-language disaster NLP benchmark exists for this region.

ID	Evidence
Q90	"Although WXIMPACTBENCH provides valuable insights (e.g., exhibit the strengths and weaknesses of various society impact understanding) about evaluating LLMs on disruptive weather, it may have potential biases in underrepresented historical events and linguistic variations." (p.9)
Q97	"Our primary data source is a corpus of three digitized newspapers (La Presse, La Patrie and Montreal Gazette), obtained through collaboration with the McGill University Library and Archives and the Bibliothèque nationale du Québec." (p.9)
WEB-7	Rio Platense Spanish is the predominant variety in Argentina/Uruguay extending to Paraguay and border regions (https://en.wikipedia.org/wiki/Rioplatense_Spanish)
WEB-9	No NLP tools handle indigenous-language lexical borrowing within Spanish disaster text (https://arxiv.org/html/2501.09943v2)
WEB-10	No mixed-language disaster classification tools exist for Guaraní/Quechua/Wichí lexical borrowing within Spanish (https://arxiv.org/html/2404.05365v1)

Strengths

- Two temporal periods (historical 1880s–1890s; modern 1990s–2000s) provide some intra-corpus register variation [Q13, D3, D16]
- Some non-Canadian wire-service content (China, Indonesia, US, Europe) appears in the labeled data [D7, D8, D16, D19], showing the corpus is not strictly Canadian — though still not Southern Cone

ID	Evidence
Q13	"We extract a sample of articles from two time periods, which cover linguistic and social changes across different eras and increase linguistic complexity due to the different descriptions of weather events in different times (Campbell, 2013)." (p.2)
D3	by six o'clock the electric car services in all the important points west of Toronto had been completely paralyzed — 1894 blizzard article: infrastructure disruption to Canadian railway/electric car — exemplary on-target sample
D7	Almost 4 million people across China had been cut off by flood waters, 810,000 homes have collapsed and 2.8 million homes have been damaged — 1996 Yangtze River flood report from China — not Canadian, annotated for infrastructure
D8	The Jakarta Post reported that hundreds of houses, schools and other buildings sustained damage when one of the twin peaks of the volcano exploded. At least 20 people sustained minor injuries — Indonesian volcanic eruption and Washington state earthquake — non-Canadian, non-weather-disaster content labeled for infrastructure and health
D16	More than 292,000 people have been evacuated in the mountainous region along the Yangtze River, with more than 100,000 homes damaged and crops on about 175,000 hectares of farmland destroyed — 2007 European/Chinese/Pakistani multi-location disaster report from Montreal Gazette — agricultural and health labeled, but not infrastructural despite 100k homes damaged

D19	With about half the original 1.6-million-hectare swamp filled for development or drained for agriculture, the park includes about 202,000 hectares of marsh experts liken the Everglades to a seriously ill patient — 1990 Everglades drought/ecosystem article — ecological only; US content
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Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — deployment inputs require knowledge of Rio Platense Spanish dialectal features, Guaraní/Quechua/Wichí toponyms (Pilcomayo, Bermejo, Tartagal, Yacuiba), and Paraguayan Jopara mixed register [WEB-7, WEB-9]; benchmark provides none of this [D1–D22 all English].

IC-2: No — benchmark content reflects North American journalistic culture (e.g., Montreal municipal snow removal [D10], US Everglades policy [D19]) rather than Southern Cone disaster-response culture [Q97].

IC-3: Yes — content frequently requires understanding of Canadian/US institutional context (Toronto electric car services [D3], Quebec ice storm [D9], US Supreme Court [D22], Berlin Wall anniversary [D5]) that does not transfer to Gran Chaco deployment.

IC-4: Regional annotator recruitment has not occurred; no AFE/SINAGIR, SEN, VIDECI, or Gendarmería personnel were involved in benchmark construction [Q110].

IC-5: Critical content validity violation: zero linguistic, geographic, or cultural overlap between benchmark inputs and deployment input distribution; results from this benchmark cannot be transferred to predict performance on Gran Chaco Spanish social media.

ID	Evidence
Q90	“Although WXIMPACTBENCH provides valuable insights (e.g., exhibit the strengths and weaknesses of various society impact understanding) about evaluating LLMs on disruptive weather, it may have potential biases in underrepresented historical events and linguistic variations.” (p.9)
Q97	“Our primary data source is a corpus of three digitized newspapers (La Presse, La Patrie and Montreal Gazette), obtained through collaboration with the McGill University Library and Archives and the Bibliothèque nationale du Québec.” (p.9)
Q110	“The annotation process was conducted by members of a research group specializing in uncovering the history of a region's climate change through the regional historical weather records.” (p.15)
D1	John's Cloudy -4 -4 United States Max Min Atlanta Cloudy 17 4 Boston Cloudy 3 -6 Chicago Cloudy 2 2 Dallas Thunderstorms 11 2 — Weather table/tabular data with no narrative — all-zero label, non-event content
D2	But the real beneficiaries of the work-at-home trend are the employees themselves... She takes a turn at the table in a lay way (5) 29, A good one is of assistance (9) — Crossword puzzle clues and work-from-home editorial — zero weather impact content
D3	by six o'clock the electric car services in all the important points west of Toronto had been completely paralyzed — 1894 blizzard article: infrastructure disruption to Canadian railway/electric car — exemplary on-target sample
D4	Police say the aggravated assault charge against Clarkson involves a February attack on Boland in which the celebrity photographer's face and chest were splashed with concentrated sulphuric acid. — Crime report with no weather content — all-zero label
D5	The decision yesterday followed a 16-hour overnight meeting in Berlin, where debate over how much to give away was sometimes interrupted by the sounds of revellers celebrating the first anniversary of the opening of the Berlin Wall. — Political news about German unification — no weather content

D6	A poor lady passenger was dashed to leeward and had her skull fractured. She was landed this morning in a dying state with a coffined child that succumbed last night to its sufferings. — Storm narrative with human casualty but annotated all-zero — potential label discrepancy
D7	Almost 4 million people across China had been cut off by flood waters, 810,000 homes have collapsed and 2.8 million homes have been damaged — 1996 Yangtze River flood report from China — not Canadian, annotated for infrastructure
D8	The Jakarta Post reported that hundreds of houses, schools and other buildings sustained damage when one of the twin peaks of the volcano exploded. At least 20 people sustained minor injuries — Indonesian volcanic eruption and Washington state earthquake — non-Canadian, non-weather-disaster content labeled for infrastructure and health
D9	residents of southwestern Quebec have been plunged back into the deep freeze Overnight lows in Montreal were expected to drop to minus-16 Celsius last night and tumble as far as minus-30C on the South Shore, where more than one million people struggle on without power or heat — 1998 Quebec ice storm aftermath — one million without power/heat, annotated all-zero
D10	Residents of the locality were put to great inconvenience by the huge piles of snow now accumulated on the streets... Already \$30,000 has been expended in removing it — 1887 Montreal snow removal municipal debate — multi-label (infra+pol+fin) correctly labeled
D11	-2 City res y 5000 5 5 5 Aieaxavn 8000 50 50 50 Ckearcdnf 835 825 25 JS 1 Almaden 1000 35 35 35 — Stock ticker/financial table — all-zero label, clearly off-topic content
D12	The walkout will interrupt regular garbage collection and will affect street and sidewalk cleanup if it snows. Unless nine centimetres or more of snow falls during the walkout, blue-collar workers aren't obliged to clean the streets and sidewalks. — Labor dispute/garbage collection strike — weather mentioned incidentally, all-zero label
D13	Penn refuses to play up the easy nebbish... he does not wear glasses, flood pants or a bow tie — Film review with only figurative use of "flood" — all-zero label, entirely off-topic
D14	in Richmond, the storm so long expected has broken forth in all its fury. Great flashes of blinding lightning intermingled with the grand roar of the thunder from on high — Fictional storm description used metaphorically in a society/romance narrative — all-zero label
D15	the agency said Christian Lambrechts... said this would expose more of the sea's surface to sunlight, rather than reflect it, contributing to continued and accelerated warming The peninsula - the tongue of land that juts toward South America - has been hit by greater warming than almost any other area on Earth — Antarctic/climate change science brief — no direct disaster impact, all-zero label
D16	More than 292,000 people have been evacuated in the mountainous region along the Yangtze River, with more than 100,000 homes damaged and crops on about 175,000 hectares of farmland destroyed — 2007 European/Chinese/Pakistani multi-location disaster report from Montreal Gazette — agricultural and health labeled, but not infrastructural despite 100k homes damaged
D17	01 SECOND RACE: Trot, I Mile Purse: 11,000 5 Prime Saliora (M. Lalonde) 9.70 5.10 5.70 3 Bidou Fio (R. SimarrJ) 5.30 4.40 — Horse racing results table with unrelated editorial on optometrists' ad — off-topic noise
D18	A line of freak thunderstorms rumbling across the northeastern Mediterranean coast from Spain and southern France to the principality of Monaco brought two days of disastrous flash flooding... Raging waters uprooted trees and swept away cars as river levels rose by as much as 3 metres — 1992 Mediterranean flash flood — political and ecological but not infrastructural ("swept away cars," "uprooted trees") — label boundary question
D19	With about half the original 1.6-million-hectare swamp filled for development or drained for agriculture, the park includes about 202,000 hectares of marsh experts liken the Everglades to a seriously ill patient — 1990 Everglades drought/ecosystem article — ecological only; US content
D20	July 7th they encountered a heavy gale from the west, and had great difficulty in keeping the boat free, the sea continually breaking on board... After the accident they suffered severely from the cold. — 1896 Atlantic rowing voyage encounter with gale — human suffering described but labeled all-zero
D21	TRAFFIC IS PARALYZED In Western Canadian Cities, and at Many Points In the United States Disasters In England. — 1894 blizzard chunk (segmented from Ex. 3) — infrastructure label carried over to chunk
D22	Peter Russell, a law professor at the University of Toronto, said the Supreme Court will benefit from Binnie's solid experience in constitutional and international affairs. — Legal/judicial appointment news — entirely off-topic
WEB-5	Cross-border displacement and indigenous community evacuation are primary Gran Chaco disaster consequences (10,000+ Wichi/Toba/Chorote/Tapiete evacuated in 2018) (https://www.iadb.org/en/staying-ahead-gran-chacos-floods)
WEB-7	Rio Platense Spanish is the predominant variety in Argentina/Uruguay extending to Paraguay and border regions (https://en.wikipedia.org/wiki/Rioplatense_Spanish)
WEB-9	No NLP tools handle indigenous-language lexical borrowing within Spanish disaster text (https://arxiv.org/html/2501.09943v2)
WEB-11	RoBERTuito (500M+ Spanish tweets) is the most relevant existing model for the deployment's input register, but is not disaster-domain-specific and not regionally evaluated (https://aclanthology.org/2022.lrec-1.785/)

Information Gaps

- No quantitative characterization of actual Gran Chaco disaster social media token-length and lexical-borrowing distribution exists; would require corpus collection from regional platforms

Requires Expert Verification

- Regional annotators (AFE/SINAGIR coordinators, SEN, VIDECl personnel) should validate which proportion of routine deployment inputs contain indigenous-language lexical borrowing requiring specialized handling
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Remediation

Gap 1: Zero Spanish-language, social-media, or Southern Cone content; no indigenous-language lexical-borrowing exemplars.

Recommendation: Construct a Gran Chaco Spanish-language evaluation corpus from Twitter/X, Facebook community groups, WhatsApp broadcast channels, and local news archives spanning Salta/Formosa/Chaco/Boquerón/Alto Paraguay/Presidente Hayes/Tarija; include an indigenous-toponym/lexical-borrowing test slice.